

Functional Inequalities for Probability Measures

Leopold Wuhan Zhou
School of EECS, Peking University

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Abstract

This note is based on Joe Jackson's lecture in 2026 summer at University of Chicago. We introduce some functional inequalities for probability measures on Euclidean space, such as the Poincare inequality and the log-Sobolev inequality. We also sketch some applications to concentration of measure and the long-time behavior of diffusion processes. Finally, we prove these inequalities in the Gaussian case, and show that some broader classes of measures also satisfy these inequalities.

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1 Poincaré Inequality

Let $\mathcal{P}(\mathbb{R}^n)$ denote the set of Borel probability measures on $\mathbb{R}^n$. Given a random vector X in $\mathbb{R}^n$, we write $X \sim \mu$ to mean that X has distribution μ , i.e.

$$\mu(A) = \mathbb{P}\{X \in A\}$$

for every Borel set $A \subseteq \mathbb{R}^n$. If X has probability density function $p : \mathbb{R}^n \rightarrow [0, +\infty)$, then

$$\mu(A) = \int_A p(x) \, dx.$$

For a measurable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, define

$$\begin{aligned} \text{Var}_\mu(f) &= \int_{\mathbb{R}^n} f^2 \, d\mu - \left(\int_{\mathbb{R}^n} f \, d\mu \right)^2 \\ &= \int_{\mathbb{R}^n} \left(f(x) - \int_{\mathbb{R}^n} f \, d\mu \right)^2 \, d\mu(x). \end{aligned}$$

If $X \sim \mu$, then $\text{Var}_\mu(f) = \text{Var}(f(X))$.

Definition 1.1 (Poincaré inequality). A probability measure $\mu \in \mathcal{P}(\mathbb{R}^n)$ satisfies a *Poincaré inequality* if there exists a constant $C_{\text{PI}} > 0$ such that

$$\text{Var}_\mu(f) \leq C_{\text{PI}}(\mu) \int_{\mathbb{R}^n} |\nabla f(x)|^2 \, d\mu(x), \quad \forall f \in C_c^\infty(\mathbb{R}^n).$$

The best possible constant is denoted by $C_{\text{PI}}(\mu)$.

Example 1.2. In dimension one, let X be the random variable satisfying

$$\mathbb{P}\{X = 1\} = \mathbb{P}\{X = -1\} = \frac{1}{2}.$$

Then $X \sim \mu$, where

$$\mu = \frac{1}{2}\delta_1 + \frac{1}{2}\delta_{-1}.$$

This is a simple discrete probability measure. The main definitions above are written for measures on Euclidean spaces, and for absolutely continuous measures the gradient term has the usual differential interpretation.

A fundamental question in probability is the following. Given independent random variables

$$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mu,$$

and a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, set

$$Z = f(X_1, \dots, X_n).$$

Under what conditions on μ and f does Z concentrate around its mean? Equivalently, when is

$$\mathbb{P}\{|Z - \mathbb{E}Z| > t\}$$

small? If f is linear, this is classical. For example,

$$f(X_1, \dots, X_n) = \frac{1}{n} \sum_{i=1}^n X_i$$

is controlled by the usual laws of large numbers and concentration bounds.

The key point is that the Poincaré inequality tensorizes.

Proposition 1.3 (Tensorization of the Poincaré inequality). *Let $\mu \in \mathcal{P}(\mathbb{R}^n)$ and $\nu \in \mathcal{P}(\mathbb{R}^d)$. Their product measure $\mu \otimes \nu \in \mathcal{P}(\mathbb{R}^{n+d})$ is defined by*

$$(\mu \otimes \nu)(dx, dy) = \mu(dx)\nu(dy).$$

Equivalently, if $X \sim \mu$, $Y \sim \nu$, and X, Y are independent, then

$$(X, Y) \sim \mu \otimes \nu.$$

If μ and ν both satisfy Poincaré inequalities, then $\mu \otimes \nu$ also satisfies a Poincaré inequality. More precisely,

$$C_{\text{PI}}(\mu \otimes \nu) = \max\{C_{\text{PI}}(\mu), C_{\text{PI}}(\nu)\}.$$

In particular,

$$C_{\text{PI}}(\mu^{\otimes n}) = C_{\text{PI}}(\mu).$$

Therefore, if μ satisfies a Poincaré inequality and $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mu$, then for every smooth $f : \mathbb{R}^n \rightarrow \mathbb{R}$,

$$\text{Var}(f(X_1, \dots, X_n)) \leq C_{\text{PI}}(\mu) \mathbb{E}[|\nabla f(X_1, \dots, X_n)|^2].$$

By Chebyshev's inequality,

$$\mathbb{P}\{|f(X_1, \dots, X_n) - \mathbb{E}f(X_1, \dots, X_n)| > t\} \leq \frac{C_{\text{PI}}(\mu) \mathbb{E}|\nabla f(X_1, \dots, X_n)|^2}{t^2}.$$

For the simplest example

$$f(X_1, \dots, X_n) = \frac{1}{n} \sum_{i=1}^n X_i,$$

we have

$$|\nabla f|^2 = \sum_{i=1}^n \left(\frac{1}{n}\right)^2 = \frac{1}{n}.$$

Thus the Poincaré inequality gives the polynomial concentration estimate

$$\mathbb{P}\left\{\left|\frac{1}{n} \sum_{i=1}^n X_i - \frac{1}{n} \sum_{i=1}^n \mathbb{E}X_i\right| > t\right\} \leq \frac{C_{\text{PI}}(\mu)}{nt^2}.$$

Let $\gamma \in \mathcal{P}(\mathbb{R})$ be the standard Gaussian measure,

$$\gamma(dx) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx,$$

and let

$$\gamma_n = \gamma^{\otimes n}$$

be the standard Gaussian measure on $\mathbb{R}^n$.

Theorem 1.4. *The standard Gaussian measure satisfies*

$$C_{\text{PI}}(\gamma) = 1.$$

Consequently, by tensorization, $C_{\text{PI}}(\gamma_n) = 1$ for every n .

Proof. Let $(B_t)_{0 \leq t \leq 1}$ be a one-dimensional Brownian motion. In particular,

$$B_1 \sim \gamma.$$

For a smooth compactly supported function f , define the martingale

$$M_t = \mathbb{E}[f(B_1) \mid \mathcal{F}_t^B],$$

where $(\mathcal{F}_t^B)$ is the filtration generated by the Brownian motion. Notice that

$$M_0 = \mathbb{E}f(B_1), \quad M_1 = f(B_1).$$

Hence

$$\text{Var}_\gamma(f) = \mathbb{E}[(M_1 - M_0)^2].$$

By the Feynman–Kac formula,

$$M_t = u(t, B_t),$$

where u solves the backward heat equation

$$\begin{cases} (\partial_t + \frac{1}{2}\partial_{xx}) u(t, x) = 0, & (t, x) \in (0, 1) \times \mathbb{R}, \\ u(1, x) = f(x), & x \in \mathbb{R}. \end{cases}$$

Applying Ito's formula gives

$$\begin{aligned} dM_t &= du(t, B_t) \\ &= \left(\partial_t u + \frac{1}{2} \partial_{xx} u \right) (t, B_t) dt + \partial_x u(t, B_t) dB_t \\ &= \partial_x u(t, B_t) dB_t. \end{aligned}$$

Therefore

$$dM_t^2 = 2M_t dM_t + |\partial_x u(t, B_t)|^2 dt,$$

and hence

$$\frac{d}{dt} \mathbb{E}[M_t^2] = \mathbb{E}|\partial_x u(t, B_t)|^2.$$

It follows that

$$\text{Var}_\gamma(f) = \int_0^1 \mathbb{E}|\partial_x u(t, B_t)|^2 dt.$$

Finally, the process $(\partial_x u(t, B_t))_{0 \leq t \leq 1}$ is also a martingale. Therefore its second moment is increasing in t , and

$$\mathbb{E}|\partial_x u(t, B_t)|^2 \leq \mathbb{E}|\partial_x u(1, B_1)|^2 = \mathbb{E}|f'(B_1)|^2.$$

Thus

$$\text{Var}_\gamma(f) \leq \int_0^1 \mathbb{E}|f'(B_1)|^2 dt = \mathbb{E}|f'(B_1)|^2 = \int_{\mathbb{R}} |f'(x)|^2 d\gamma(x).$$

This proves $C_{\text{PI}}(\gamma) \leq 1$. Equality follows by testing the inequality on $f(x) = x$, for which

$$\text{Var}_\gamma(x) = 1, \quad \int_{\mathbb{R}} |f'(x)|^2 d\gamma(x) = 1.$$

Therefore $C_{\text{PI}}(\gamma) = 1$. □

2 Log-Sobolev Inequality

For a nonnegative function $f : \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$, define its entropy with respect to $\mu \in \mathcal{P}(\mathbb{R}^n)$ by

$$\begin{aligned} \text{Ent}_\mu(f) &= \int f \log f \, d\mu - \left(\int f \, d\mu \right) \log \left(\int f \, d\mu \right) \\ &= \int \varphi(f) \, d\mu - \varphi \left(\int f \, d\mu \right), \end{aligned}$$

where

$$\varphi(z) = \begin{cases} z \log z, & z > 0, \\ 0, & z = 0. \end{cases}$$

Definition 2.1 (Log-Sobolev inequality). A probability measure $\mu \in \mathcal{P}(\mathbb{R}^n)$ satisfies a *log-Sobolev inequality* with constant $C_{\text{LSI}}(\mu)$ if

$$\text{Ent}_\mu(f^2) \leq 2C_{\text{LSI}}(\mu) \int_{\mathbb{R}^n} |\nabla f|^2 \, d\mu$$

for all sufficiently regular functions f .

Remark 2.2. The log-Sobolev inequality implies the Poincare inequality. More precisely,

$$C_{\text{PI}}(\mu) \leq C_{\text{LSI}}(\mu).$$

This can be seen by applying the log-Sobolev inequality to $1 + \varepsilon f$ and taking the limit as $\varepsilon \rightarrow 0$.

There is also an equivalent information-theoretic formulation. If $\nu \ll \mu$ and

$$h = \frac{d\nu}{d\mu},$$

define the relative entropy and relative Fisher information by

$$H(\nu \mid \mu) = \int \log \left(\frac{d\nu}{d\mu} \right) \, d\nu = \int h \log h \, d\mu,$$

and

$$I(\nu \mid \mu) = \int \left| \nabla \log \left(\frac{d\nu}{d\mu} \right) \right|^2 \, d\nu = \int \frac{|\nabla h|^2}{h} \, d\mu.$$

Then the log-Sobolev inequality is equivalent to

$$H(\nu \mid \mu) \leq \frac{C_{\text{LSI}}(\mu)}{2} I(\nu \mid \mu).$$

Indeed, one applies the functional form to $f = \sqrt{h}$.

The log-Sobolev inequality also tensorizes. This comes from the tensorization of entropy. Thus, if μ satisfies a log-Sobolev inequality, then

$$C_{\text{LSI}}(\mu^{\otimes n}) = C_{\text{LSI}}(\mu).$$

Let Z be a real-valued random variable with $\mathbb{E}Z = 0$. Define the log-Laplace transform

$$\psi(\lambda) = \psi_Z(\lambda) = \log \mathbb{E}e^{\lambda Z}.$$

Equivalently, if Z has law μ , then

$$\psi_Z(\lambda) = \log \int_{\mathbb{R}} e^{\lambda z} d\mu(z).$$

Proposition 2.3 (Chernoff bound). *Suppose that, for some $\xi > 0$,*

$$\psi(\lambda) \leq \frac{\xi^2 \lambda^2}{2}, \quad \forall \lambda \in \mathbb{R}.$$

Then

$$\mathbb{P}\{Z > t\} \leq e^{-t^2/(2\xi^2)}.$$

By applying the same argument to $-Z$, one also obtains the two-sided estimate

$$\mathbb{P}\{|Z| > t\} \leq 2e^{-t^2/(2\xi^2)}.$$

Proof. For any $\lambda > 0$,

$$\begin{aligned} \mathbb{P}\{Z > t\} &= \mathbb{P}\{e^{\lambda Z} > e^{\lambda t}\} \\ &\leq e^{-\lambda t} \mathbb{E}e^{\lambda Z} \\ &= e^{-\lambda t + \psi(\lambda)} \\ &\leq \exp\left(-\lambda t + \frac{\xi^2 \lambda^2}{2}\right). \end{aligned}$$

Optimizing in λ gives $\lambda = t/\xi^2$, and therefore

$$\mathbb{P}\{Z > t\} \leq e^{-t^2/(2\xi^2)}.$$

□

The strategy is to use the log-Sobolev inequality to obtain a Chernoff bound for

$$Z = f(X), \quad X \sim \mu.$$

Proposition 2.4 (Concentration from the log-Sobolev inequality). *Assume that $\mu \in \mathcal{P}(\mathbb{R}^n)$ satisfies a log-Sobolev inequality with constant $C_{\text{LSI}}(\mu)$. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be L -Lipschitz. If $X \sim \mu$, then*

$$\mathbb{P}\{|f(X) - \mathbb{E}f(X)| > t\} \leq 2 \exp\left(-\frac{t^2}{2C_{\text{LSI}}(\mu)L^2}\right).$$

Proof sketch. Set

$$Z = f(X) - \mathbb{E}f(X).$$

For notational simplicity, assume $\mathbb{E}f(X) = 0$, so $Z = f(X)$. We want an upper bound on

$$\psi(\lambda) = \log \mathbb{E}e^{\lambda f(X)}.$$

Note that

$$\psi'(\lambda) = \frac{\mathbb{E}[f(X)e^{\lambda f(X)}]}{\mathbb{E}e^{\lambda f(X)}}, \quad \psi'(0) = \mathbb{E}f(X) = 0.$$

The key computation is

$$\begin{aligned} \frac{d}{d\lambda} \left(\frac{\psi(\lambda)}{\lambda} \right) &= \frac{\lambda\psi'(\lambda) - \psi(\lambda)}{\lambda^2} \\ &= \frac{\lambda\mathbb{E}[fe^{\lambda f}] - \mathbb{E}(e^{\lambda f}) \log \mathbb{E}(e^{\lambda f})}{\lambda^2 \mathbb{E}(e^{\lambda f})} \\ &= \frac{\text{Ent}_\mu(e^{\lambda f})}{\lambda^2 \mathbb{E}(e^{\lambda f})}. \end{aligned}$$

Apply the log-Sobolev inequality to

$$g = e^{\lambda f/2}.$$

Since $g^2 = e^{\lambda f}$,

$$\begin{aligned} \text{Ent}_\mu(e^{\lambda f}) &= \text{Ent}_\mu(g^2) \\ &\leq 2C_{\text{LSI}}(\mu) \int |\nabla g|^2 d\mu \\ &= 2C_{\text{LSI}}(\mu) \int \left| \frac{\lambda}{2} e^{\lambda f/2} \nabla f \right|^2 d\mu \\ &\leq \frac{\lambda^2 C_{\text{LSI}}(\mu) L^2}{2} \int e^{\lambda f} d\mu. \end{aligned}$$

Therefore

$$\frac{d}{d\lambda} \left(\frac{\psi(\lambda)}{\lambda} \right) \leq \frac{C_{\text{LSI}}(\mu) L^2}{2}.$$

Since $\lim_{\lambda \rightarrow 0} \psi(\lambda)/\lambda = \mathbb{E}f(X) = 0$, we get

$$\psi(\lambda) \leq \frac{C_{\text{LSI}}(\mu) L^2}{2} \lambda^2.$$

The Chernoff bound then gives

$$\mathbb{P}\{f(X) - \mathbb{E}f(X) > t\} \leq \exp\left(-\frac{t^2}{2C_{\text{LSI}}(\mu)L^2}\right),$$

and applying the same argument to $-f$ gives the two-sided bound. $\square$

Corollary 2.5. *Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mu$, and assume that μ satisfies a log-Sobolev inequality. If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is L -Lipschitz, then*

$$\mathbb{P}\{|f(X_1, \dots, X_n) - \mathbb{E}f(X_1, \dots, X_n)| > t\} \leq 2 \exp\left(-\frac{t^2}{2C_{\text{LSI}}(\mu)L^2}\right),$$

where we used $C_{\text{LSI}}(\mu^{\otimes n}) = C_{\text{LSI}}(\mu)$. In particular, for

$$f(x_1, \dots, x_n) = \frac{1}{n} \sum_{i=1}^n x_i,$$

we have $L = 1/\sqrt{n}$, hence

$$\mathbb{P}\left\{\left|\frac{1}{n} \sum_{i=1}^n X_i - \mathbb{E}\frac{1}{n} \sum_{i=1}^n X_i\right| > t\right\} \leq 2 \exp\left(-\frac{nt^2}{2C_{\text{LSI}}(\mu)}\right).$$

3 Log-Concave Measures

A natural question is: which measures, other than the standard Gaussian measure, satisfy Poincare inequalities and log-Sobolev inequalities?

Definition 3.1 (Log-concave measure). A probability measure $\mu \in \mathcal{P}(\mathbb{R}^n)$ is *log-concave* if it has the form

$$\mu(\mathrm{d}x) = \frac{1}{Z} e^{-V(x)} \mathrm{d}x,$$

where $V : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex and

$$Z = \int_{\mathbb{R}^n} e^{-V(x)} \mathrm{d}x > 0.$$

It is called C -log-concave if V is C -convex, i.e.

$$x \mapsto V(x) - \frac{C}{2}|x|^2$$

is convex. If V is twice continuously differentiable, this is equivalent to

$$D^2V(x) \geq CI_n, \quad \forall x \in \mathbb{R}^n.$$

Theorem 3.2 (Bakry–Emery). *If μ is C -log-concave with $C > 0$, then*

$$C_{\text{PI}}(\mu) \leq \frac{1}{C}, \quad C_{\text{LSI}}(\mu) \leq \frac{1}{C}.$$

3.1 Pushforwards and Lipschitz Maps

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^k$ be a measurable map and let $\mu \in \mathcal{P}(\mathbb{R}^n)$. The pushforward measure $T_{\#}\mu \in \mathcal{P}(\mathbb{R}^k)$ is defined by

$$T_{\#}\mu(A) = \mu(T^{-1}(A)).$$

Equivalently, if $X \sim \mu$, then

$$T(X) \sim T_{\#}\mu.$$

The guiding question is whether Poincare inequalities or log-Sobolev inequalities for μ imply corresponding inequalities for $T_{\#}\mu$.

Assume that μ satisfies a Poincare inequality. For a smooth function $f : \mathbb{R}^k \rightarrow \mathbb{R}$,

$$\begin{aligned} \text{Var}_{T_{\#}\mu}(f) &= \text{Var}_{\mu}(f \circ T) \\ &\leq C_{\text{PI}}(\mu) \int |\nabla(f \circ T)(x)|^2 \mathrm{d}\mu(x) \\ &\leq C_{\text{PI}}(\mu) \text{Lip}(T)^2 \int |\nabla f(T(x))|^2 \mathrm{d}\mu(x) \\ &= \text{Lip}(T)^2 C_{\text{PI}}(\mu) \int |\nabla f(y)|^2 \mathrm{d}(T_{\#}\mu)(y). \end{aligned}$$

Thus we have the following result.

Proposition 3.3. *If μ satisfies a Poincare inequality and $T : \mathbb{R}^n \rightarrow \mathbb{R}^k$ is L -Lipschitz, then $T_{\#}\mu$ also satisfies a Poincare inequality, with*

$$C_{\text{PI}}(T_{\#}\mu) \leq L^2 C_{\text{PI}}(\mu).$$

The same proof, applied to the entropy functional, also gives the analogous statement for log-Sobolev inequalities:

$$C_{\text{LSI}}(T_{\#}\mu) \leq L^2 C_{\text{LSI}}(\mu).$$

The idea is now clear. Given a C -log-concave measure μ , one tries to construct a Lipschitz map

$$T : \mathbb{R}^n \rightarrow \mathbb{R}^n$$

such that

$$T_{\#}\gamma_n = \mu.$$

If T is sufficiently contracting, the functional inequalities for γ_n transfer to μ .

3.2 Optimal Transport

The relevant construction comes from optimal transport. Given $\mu, \nu \in \mathcal{P}(\mathbb{R}^n)$, the Monge problem with quadratic cost is

$$\inf_{T_{\#}\mu=\nu} \int_{\mathbb{R}^n} |T(x) - x|^2 d\mu(x).$$

Theorem 3.4 (Caffarelli contraction theorem). *Suppose the source measure is γ_n and the target measure is*

$$\mu(dx) = \frac{1}{Z} e^{-V(x)} dx,$$

where μ is C -log-concave. Then there exists an optimal transport map T^ with*

$$T_{\#}^* \gamma_n = \mu,$$

and T^ is $1/\sqrt{C}$ -Lipschitz.*

Combining this contraction estimate with the pushforward principle above gives another route to the bounds

$$C_{\text{PI}}(\mu) \leq \frac{1}{C}, \quad C_{\text{LSI}}(\mu) \leq \frac{1}{C},$$

starting from the Gaussian inequalities

$$C_{\text{PI}}(\gamma_n) = 1, \quad C_{\text{LSI}}(\gamma_n) = 1.$$

3.3 Langevin Dynamics

Fix a convex function $V : \mathbb{R}^n \rightarrow \mathbb{R}$ and consider the stochastic differential equation

$$dX_t = -\nabla V(X_t) dt + \sqrt{2} dW_t,$$

where W_t is an n -dimensional Brownian motion. The main question is: what is the long-time behavior of X_t ?

Let

$$\mu_t = \text{Law}(X_t \mid X_0 \sim \mu_0),$$

and view $(\mu_t)_{t \geq 0}$ as a curve in $\mathcal{P}(\mathbb{R}^n)$. The invariant measure is

$$\mu_\infty(\mathrm{d}x) = \frac{1}{Z} e^{-V(x)} \mathrm{d}x,$$

and one has

$$\mu_t \xrightarrow{t \rightarrow \infty} \mu_\infty.$$

A key fact is entropy dissipation:

$$\frac{\mathrm{d}}{\mathrm{d}t} H(\mu_t \mid \mu_\infty) = -I(\mu_t \mid \mu_\infty).$$

With the convention used in this note,

$$H(\nu \mid \mu_\infty) \leq \frac{C_{\text{LSI}}(\mu_\infty)}{2} I(\nu \mid \mu_\infty),$$

so

$$I(\nu \mid \mu_\infty) \geq \frac{2}{C_{\text{LSI}}(\mu_\infty)} H(\nu \mid \mu_\infty).$$

Therefore

$$\frac{\mathrm{d}}{\mathrm{d}t} H(\mu_t \mid \mu_\infty) \leq -\frac{2}{C_{\text{LSI}}(\mu_\infty)} H(\mu_t \mid \mu_\infty).$$

By Gronwall's inequality,

$$H(\mu_t \mid \mu_\infty) \leq H(\mu_0 \mid \mu_\infty) \exp\left(-\frac{2t}{C_{\text{LSI}}(\mu_\infty)}\right).$$

Equivalently, if the log-Sobolev inequality is written in rate form

$$H(\nu \mid \mu_\infty) \leq \frac{1}{2\alpha_{\text{LSI}}} I(\nu \mid \mu_\infty),$$

then

$$H(\mu_t \mid \mu_\infty) \leq H(\mu_0 \mid \mu_\infty) e^{-2\alpha_{\text{LSI}} t}.$$